

LECTURE EIGHT

Recap from last lecture

$$y_t = \beta_0 + \beta_1 \cos(2\pi f t) + \beta_2 \sin(2\pi f t) + \varepsilon_t$$

$t = 1, \dots, n$

$y_1, \dots, y_n$ $\beta_0, \beta_1, \beta_2$ main parameters $\varepsilon_t \stackrel{iid}{\sim} N(0, \sigma^2)$

$$RSS(f) = \min_{\beta_0, \beta_1, \beta_2} \sum_{t=1}^n (y_t - \beta_0 - \beta_1 \cos 2\pi f t - \beta_2 \sin 2\pi f t)^2$$

We can restrict f to $[0, 0.5]$

- Take a grid of f -values.
- Compute $RSS(f)$
- Minimize to obtain $\hat{f}$.

For efficient computation, we need a more explicit expression for $RSS(f)$.

Fourier Frequencies: $f \in [0, 0.5]$

(nf must be an integer)

$$\begin{cases} \left\{ 0, \frac{1}{n}, \frac{2}{n}, \frac{3}{n}, \dots, \frac{n-1}{2n} \right\} & \text{if } n \text{ is odd} \\ \left\{ 0, \frac{1}{n}, \frac{2}{n}, \frac{3}{n}, \dots, \frac{n/2}{n} \right\} & \text{if } n \text{ is even} \end{cases}$$

LAST CLASS, we derived

$$RSS(f) =$$

when f is a Fourier frequency $\wedge 0 < f < \frac{1}{2}$

$$\downarrow \sum_{t=1}^n (y_t - \bar{y})^2 - \frac{2}{n} \left(\sum_{t=1}^n y_t \cos 2\pi f t \right)^2 - \frac{2}{n} \left(\sum_{t=1}^n y_t \sin 2\pi f t \right)^2$$

$$\begin{aligned} y \cdot \cos &= \sum_{t=1}^n y_t \cos 2\pi f t \\ y \cdot \sin &= \sum_{t=1}^n y_t \sin 2\pi f t \end{aligned}$$

$$RSS(f) = \sum_{t=1}^n (y_t - \bar{y})^2 - \frac{2}{n} \left[(y \cdot \cos)^2 + (y \cdot \sin)^2 \right]$$

$$I(f) = \frac{1}{n} \left\{ (y \cdot \cos)^2 + (y \cdot \sin)^2 \right\} \quad \begin{array}{l} 0 < f < \frac{1}{2} \\ f \text{ is a} \\ \text{Fourier} \\ \text{frequency} \end{array}$$

$$RSS(f) = \sum_{t=1}^n (y_t - \bar{y})^2 - 2 I(f)$$

Periodogram

Minimizing $RSS(f)$ over
Fourier
frequencies

Maximizing $I(f)$
over Fourier
frequencies

$$I(f) = \frac{1}{n} \left[(y \cdot \cos)^2 + (y \cdot \sin)^2 \right]$$

$$\sum_{t=1}^n y_t e^{-2\pi i f t} = y \cdot e^{2\pi i f t}$$

$$[e^{i\theta} = \cos \theta + i \sin \theta]$$

$$I(f) = \frac{1}{n} \left| \underbrace{y \cdot e^{2\pi i f t}} \right|^2$$

Real part: $y \cdot \cos$

Imaginary part: $-y \cdot \sin$

Dot product between complex vectors:

$$a = (a_1, \dots, a_n) \quad b = (b_1, \dots, b_n)$$

$$a \cdot b = \langle a, b \rangle = \sum_{j=1}^n a_j \overline{b_j} \rightarrow \text{complex conjugate of } b_j$$

$$y \cdot e^{2\pi i f t} = \sum_{t=1}^n y_t e^{-2\pi i f t}$$

Periodogram: $I(f) = \frac{1}{n} \left| \underbrace{y \cdot e^{2\pi i f t}} \right|^2$

DFT (Discrete Fourier Transform)

Data $y_0, y_1, \dots, y_{n-1}$ (often represented by the vector y)

$$b_j = \sum_{t=0}^{n-1} y_t \exp\left(-2\pi i \frac{j}{n} t\right)$$

represents the Fourier frequency $\frac{j}{n}$

$$y \cdot \underbrace{\exp\left(2\pi i \frac{j}{n} t\right)}$$

$$y = (y_0 \dots y_{n-1})$$

$$u^j = (\exp(2\pi i \frac{j}{n} t), t=0, 1, \dots, n-1)$$

↳ Data coming from the complex sinusoid of frequency $\frac{j}{n}$.

$$b_j = y \cdot u^j = \langle y, u^j \rangle$$

Indexing starting at 0 vs 1

$$b_j = \sum_{t=0}^{n-1} y_t \exp(-2\pi i \frac{j}{n} t)$$

$$\tilde{b}_j = \sum_{t=1}^n y_t \exp(-2\pi i \frac{j}{n} t)$$

$$b_j = \text{first data point} \times e^{-2\pi i \frac{j}{n} (0)} + \text{2nd data point} \times e^{-2\pi i \frac{j}{n} (1)} + \dots + \text{tth data point} \times e^{-2\pi i \frac{j}{n} (t-1)}$$

$$\tilde{b}_j = \text{first data point} \times e^{-2\pi i \frac{j}{n} (1)} + \text{2nd data point} \times e^{-2\pi i \frac{j}{n} (2)} + \dots + \text{tth data point} \times e^{-2\pi i \frac{j}{n} t} + \dots$$

$$b_j = \tilde{b}_j \exp\left(\frac{2\pi i j}{n}\right)$$

Check then $|b_j| = |\tilde{b}_j|$

$$b_j = \sum_{t=0}^{n-1} y_t \exp\left(-2\pi i \frac{j}{n} t\right)$$

$j = 0, 1, \dots, n-1$

$(b_0, \dots, b_{n-1})$ is called the DFT of $y_0, \dots, y_{n-1}$

Note: Data $y_0, \dots, y_{n-1}$ are always real but the DFT b_j can be complex

$$I\left(\frac{j}{n}\right) = \frac{|b_j|^2}{n} \quad \text{for } 0 < \frac{j}{n} < \frac{1}{2}$$

$$\text{RSS}\left(\frac{j}{n}\right) = \sum (y_t - \bar{y})^2 - 2 I\left(\frac{j}{n}\right)$$

IMPORTANT: DFT $b_j, j=0, 1, \dots, n-1$

can be computed efficiently by the FFT algorithm.

FAST FOURIER TRANSFORM

COOLEY & TUKEY.

$$O(n \log n)$$

$n=4$ $n=2$

More on DFT

$$b_j = \sum_{t=0}^{n-1} y_t \exp(-2\pi i \frac{j}{n} t),$$

$j = 0, 1, \dots, n-1$

① $b_0 = \sum_{t=0}^{n-1} y_t$ } uninteresting
always real

② $b_j = \underbrace{\sum_{t=0}^{n-1} y_t \cos 2\pi \frac{j}{n} t}_{\text{Re}(b_j)} - i \underbrace{\sum_{t=0}^{n-1} y_t \sin 2\pi \frac{j}{n} t}_{\text{Im}(b_j)}$

Generally b_j will be complex.
(Sometimes b_j can be real e.g. n even $j = \frac{n}{2}$)

③ $b_{n-j} = \sum_{t=0}^{n-1} y_t \exp(-2\pi i \frac{n-j}{n} t)$

$$= \sum_{t=0}^{n-1} y_t \underbrace{\exp(-2\pi i t)}_{=1} \exp(2\pi i \frac{j}{n} t)$$
$$= \sum_{t=0}^{n-1} y_t \exp(2\pi i \frac{j}{n} t)$$
$$= \sum_{t=0}^{n-1} y_t \exp(-2\pi i \frac{j}{n} t)$$

$e^{i\theta} = \cos\theta + i\sin\theta$
 $e^{-i\theta} = \cos\theta - i\sin\theta$
 $e^{i\theta} = \overline{e^{-i\theta}}$

$$= \sum_{t=0}^{n-1} y_t \exp(-2\pi i \frac{j}{n} t)$$

because y_t is real

$$= b_j \quad \text{Thus} \quad b_{n-j} = \overline{b_j}$$

$$n=10$$

$$j=5$$

$$b_5 = \overline{b_5}$$

$$n=11$$

$$y_0, y_1, \dots, y_{10}$$

$$b_0, b_1, b_2, b_3, b_4, b_5, \overline{b_5}, \overline{b_4}, \overline{b_3}, \overline{b_2}, \overline{b_1}$$

↑
real

complex

$$I(\frac{j}{n}) = \frac{|b_j|^2}{n}$$

$$I(\frac{6}{11}) = I(\frac{5}{11})$$

$$n=10$$

$$b_0, b_1, b_2, b_3, b_4, b_5, \overline{b_4}, \overline{b_3}, \overline{b_2}, \overline{b_1}$$

↑
real

(3) You can recover the data from the DFT.

$$b_0, b_1, b_2, \dots, b_{n-1}$$

$$y_t = \frac{1}{n} \sum_{j=0}^{n-1} b_j \exp(2\pi i \frac{j}{n} t) \rightarrow \text{Inverse DFT}$$

$$b_j = \sum_{t=0}^{n-1} y_t \exp(-2\pi i \frac{j}{n} t) \rightarrow \text{Defn of DFT}$$

Proof of IDFT

We use orthogonality of complex sinusoidal vectors:

$$u^j = \left(\exp(2\pi i \frac{j}{n} t), t = 0, 1, \dots, n-1 \right) \\ = \left(1, \exp(2\pi i \frac{j}{n} (1)), \exp(2\pi i \frac{j}{n} (2)), \dots, \exp(2\pi i \frac{j}{n} (n-1)) \right)$$

$$b_j = \langle y, u^j \rangle \rightarrow \text{Defn of DFT}$$

FACT: $u^0, u^1, \dots, u^{n-1}$ satisfy:

$$\langle u^j, u^k \rangle = \begin{cases} 0 & \text{if } j \neq k \\ n & \text{if } j = k \end{cases}$$

$$\exp(2\pi i f t)$$

$u^0, u^1, \dots, u^{n-1} \rightarrow$ vectors of length n orthogonal & constant length

$\Downarrow$
they form a basis for all vectors of length n .

$\Downarrow$
every vector can be written as a linear combination of $u^0, u^1, \dots, u^{n-1}$

$$y = a_0 u^0 + a_1 u^1 + \dots + a_{n-1} u^{n-1}$$

Take inner product on both sides with u^j .

$$\langle y, u^j \rangle = a_0 \langle u^0, u^j \rangle + a_1 \langle u^1, u^j \rangle + \dots + a_{n-1} \langle u^{n-1}, u^j \rangle$$

$$= a_j \langle u^j, u^j \rangle = a_j n$$

$$\Rightarrow a_j = \frac{1}{n} \langle y, u^j \rangle = \frac{b_j}{n}$$

We proved:

$$y = \frac{b_0}{n} u^0 + \frac{b_1}{n} u^1 + \dots + \frac{b_{n-1}}{n} u^{n-1}$$

$$\Rightarrow y_t = \frac{b_0}{n} u_t^0 + \frac{b_1}{n} u_t^1 + \dots + \frac{b_{n-1}}{n} u_t^{n-1}$$

$$y_t = \frac{1}{n} \sum_{j=0}^{n-1} b_j \exp\left(2\pi i \frac{j}{n} t\right)$$

→ Inverse DFT

Data
TIME
DOMAIN
ANALYSIS
 $y(t)$

→ DFT
Modeling
 b_j

FREQUENCY
DOMAIN
ANALYSIS